

exoatlas: Friendly Python Code for Exoplanet Populations

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Summary

Planets are complicated. Understanding how they work requires connecting individual objects to the context of broader populations. Exoplanets are easier to picture next to their closest Solar System archetypes, and planets in the Solar System are richer when seen alongside a growing community of known exoplanets in the Milky Way. The exoatlas toolkit provides a friendly Python interface for retrieving and working with populations of planets, aiming to simplify the process of placing worlds in context.

Statement of need

exoatlas was designed to meet a need among both researchers and educators for an intuitive Python tool to access planet populations. Particularly in working with students and junior scientists, for whom easy avenues for exploration and play would have particular benefit, significant barriers often frustrate attempts to perform the following tasks:

- retrieving basic properties for exoplanets and Solar System objects
- calculating derived planet quantities with propagated uncertainties
- comparing individual exoplanets to relevant comparison samples
- planning future telescope observations of known exoplanet systems
- creating beautiful and up-to-date planet data visualizations

Online planetary data archives merge incredible curatorial efforts with powerful tools for data access and visualization. These archives include the NASA Exoplanet Archive (Christiansen et al., 2025), exoplanet.eu (exoplanet.eu, 2026), exo.MAST (STScI, 2026), the University of Geneva Data Analysis Center for Exoplanets (Buchschacher & Alesina, 2019), TEPcat (Southworth, 2011), and the Open Exoplanet Catalog (Rein, 2012) for exoplanets, and the JPL Solar System Dynamics (JPL, 2026), IAU Minor Planet Center (IAU, 2026), and the NSSDCA Planetary Facts sheets (Williams, 2026) for Solar System objects. Through web portals, many of these archives enable the user to find or quickly create up-to-date plots of planetary properties. Through downloadable tables and/or application programming interfaces (APIs), they can provide raw or curated planetary properties with clear citations to the scientific literature. Drawing from these and other archives, tools like NASA's Eyes (NASA, 2026) and Stellarium (Zotti et al., 2021) allow the easy visualization of planetary orbits, and tools like Tapir (Jensen, 2013), the Exoplanet Archive Transit and Ephemeris Service (Christiansen et al., 2025), and VarAstro (Society, 2026) help plan new exoplanet observations.

41 exoatlas does not intend to replace any of these important archival efforts, web visualization
 42 portals, or planning tools. The main gap that exoatlas aims to address is to support new
 43 calculations and creative visualizations by users who are fluent in Python but uninterested
 44 in carefully parsing complicated tables, calculating replacements for missing quantities, or
 45 manually propagating uncertainties into derived quantities. By trying to do a good job of these
 46 steps behind the scenes, exoatlas aims to provide an approachable interface for exploratory
 47 analysis, centering a strong focus on scientific storytelling and illuminating visualizations. The
 48 target audience is the scientist, educator, or student who wants to curiously explore one step
 49 beyond a web plot, to playfully interact with exoplanet and Solar System data in Python, to
 50 fussily adjust how data are filtered or plotted, and/or to easily integrate new planet discoveries
 51 into their workflow.

52 Mapping populations with exoatlas

53 The user interface for exoatlas centers around the Python Population class, with each
 54 Population object containing a standardized table of planet properties and methods for
 55 interacting with that table. exoatlas makes extensive use of astropy (Astropy Collaboration,
 56 2026) to be as familiar as possible for modern astronomers, and it is thoroughly documented
 57 with astronomical audiences in mind.

- 58 ■ To retrieve planet data, exoatlas provides Population objects that automatically access
 59 archive data for exoplanets as well as Solar System major planets, minor planets, and
 60 moons. Exoplanet data come from the NASA Exoplanet Archive (Christiansen et al.,
 61 2025) API, and Solar System data come from the JPL Solar System Dynamics (JPL,
 62 2026) API or small reformatted data tables included in the package repository itself.
 63 Whatever the original data source, all Population objects act similarly and have uniform
 64 nomenclature for accessing data columns. Planet quantities all have physical units
 65 attached with astropy.units to facilitate unit conversions and minimize conceptual
 66 errors. New Population objects can also be created from astropy.table tables, enabling
 67 custom datasets to be included.
- 68 ■ To calculate derived quantities for planets, a set of default methods are included within
 69 the core Population definition, or users may attach their own calculation methods. If
 70 data for a quantity is missing, calculations can be used to swap in alternate estimates;
 71 for example, a planet's semimajor axis a will attempt first to pull from the original
 72 data table, and then second to calculate a from the planet's period P and the star's
 73 mass M_* assuming Newton's Version of Kepler's Third Law $P^2 = 4\pi^2 a^3 / GM_*$, and
 74 then third to calculate from a transit-derived ratio a/R_* . Uncertainties on all derived
 75 quantities can be numerically propagated using the astropy.uncertainty framework,
 76 where distributions of samples are generated for each original table quantity, carried
 77 through calculations, and then used to estimate confidence intervals.
- 78 ■ To extract sets of planets meeting particular criteria, Population objects can be indexed,
 79 sliced, and masked to generate new smaller Population objects. Coupling familiar
 80 array or table operations into the creation of subpopulations enables nuanced filtering of
 81 datasets based on any combination of original archival table quantities, derived quantities,
 82 and/or quantity uncertainties.
- 83 ■ To plan telescope observations of known exoplanet systems, exoatlas can estimate the
 84 signal-to-noise (S/N) ratio achievable for exoplanet observables. As derived quantities,
 85 all S/N estimates can include propagated uncertainties, to help avoid biasing target
 86 samples toward systems with larger uncertainties. Using astrolplan (Morris et al., 2018)
 87 and astropy.coordinates (Astropy Collaboration, 2026), exoatlas can determine local
 88 altitudes and azimuths for all elements in a Population, as well as upcoming opportunities
 89 to observe transits from ground-based telescopes (when the star is above the horizon, the
 90 Sun is down, and the planet is passing in front of the star). Including basic observation
 91 planning tools enables a workflow where targets can be filtered both by the estimated
 92 detectability of a signal and by whether a telescope can actually point at the system.

- To make explanatory illustrations of planet data, exoatlas includes a visual language to build up population comparison plots. The core elements of this visual language are the Plottable (a quantity that should be represented with certain scaling, limits, and labels), the Map (a panel expressing plottable quantities with position, size, or color), and the Gallery (a collection of maps with linked axes and/or datasets). Preset visualizations built from this language can offer quick contextual reference for a planet's fundamental properties, as in Figure 1, which was generated with only six lines of code:

```
from exoatlas import *
from exoatlas.visualizations import *

e = TransitingExoplanets()
s = SolarSystem()
h = e["HD209458b"]

PlanetGallery().build([e, s, h])
```

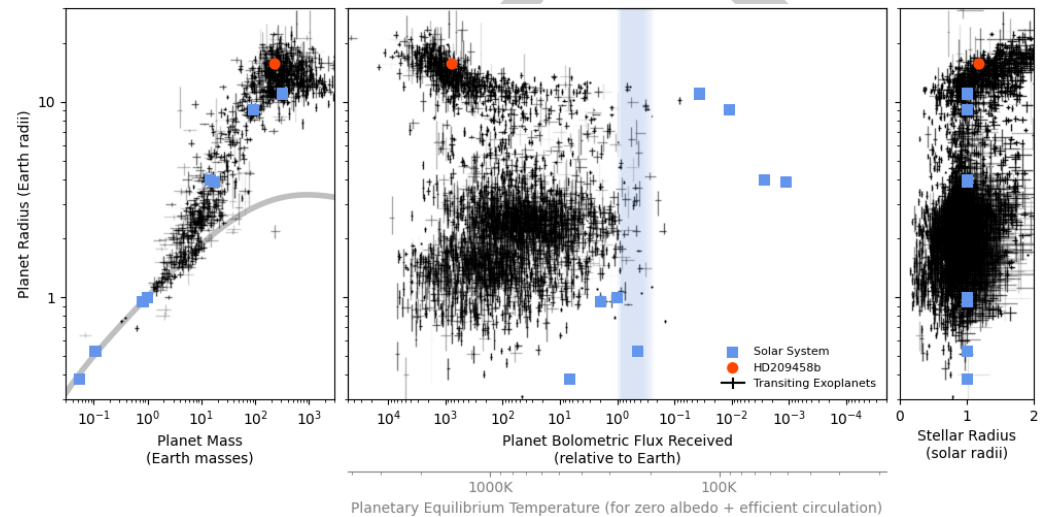


Figure 1: Example exoatlas visualization placing the first discovered transiting exoplanet HD209458b in context with other transiting exoplanets and the eight major Solar System planets. Errorbars use a color intensity that scales inversely with quantity uncertainties, to avoid giving undue visual weight to the least precise data.

Research, teaching, and learning

exoatlas was designed to support the researcher who wants to contextualize planet populations in papers, proposals, or talks, the educator who wants help connect lessons more immediately to real exoplanet and Solar System data, and the student who simply wants to learn a little more about oodles of neat planets. All these communities are encouraged try exoatlas, to ask for help, to suggest improvements, and/or to contribute code.

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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This work made use of the following software packages: astropy (Astropy Collaboration et al., 2013, 2018, 2022; Astropy Collaboration, 2026), Jupyter (Kluyver et al., 2016; Perez & Granger, 2007), matplotlib (Hunter, 2007), pandas (McKinney, 2010; The pandas development team, 2026), python (Van Rossum & Drake, 2009), scipy (Gommers et al., 2026; Virtanen et al., 2020) ArviZ (Kumar et al., 2019; Martin et al., 2026), astroquery (Ginsburg et al., 2019, 2025), tqdm (Costa-Luis et al., 2026), and xarray (Stephan Hoyer et al., 2025; S. Hoyer & Hamman, 2017). Software citation information aggregated using The Software Citation Station (Wagg et al., 2025; Wagg & Broekgaarden, 2024).

This research has made use of data and/or services provided by the International Astronomical Union's Minor Planet Center. This research has made use of data obtained from or tools provided by the portal exoplanet.eu of The Extrasolar Planets Encyclopaedia. This research has made use of the exo.MAST service at the Mikulski Archive for Space Telescopes (MAST), operated by the Space Telescope Science Institute (STScI). STScI is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS5-26555.

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